PENETRATION TESTING REPORT

FOOTPRINTING & RECONNAISSANCE ACTIVITIES

EDUCATIONAL CYBERSECURITY PROJECT

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Programme / Batch	B084
Instructor	Waqas Karim
Project date	13 November 2026
Target supplied	google.com
Environment	Kali Linux

Prepared for educational and training documentation

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1. Liability Disclaimer & Authorisation

This document records the learning activities and outputs supplied for an educational cybersecurity exercise. The evidence provided shows public-information lookups and basic HTTP/DNS observations only. It does not demonstrate exploitation, vulnerability validation, authenticated testing, or access to Google's internal systems. The materials provided do not include written authorisation from the target owner. Therefore, this report does not claim that Google authorised testing. Any further testing must only be performed within a written, explicitly approved scope and in accordance with the target owner's rules.

2. Introduction

This report documents introductory footprinting and reconnaissance activities performed in a Kali Linux terminal against the domain google.com on 12 November 2026. The exercises used WHOIS, WhatWeb, nslookup, cURL, Wafw00f, DNSRecon, and Nmap (through Zenmap) to observe publicly available registration data, DNS records, web response characteristics, generic WAF-fingerprinting results, and basic host availability. The report is limited to the commands and outputs supplied by the trainee.

2.1 Scope and Limitations

Target entered in the commands: google.com.

Activities documented: WHOIS lookup, WhatWeb fingerprinting, DNS resolution with nslookup, HTTP response inspection with cURL, generic WAF detection with Wafw00f, DNS record enumeration with DNSRecon, and a basic Nmap host-discovery scan.

No exploitation, password attacks, vulnerability exploitation, authenticated testing, or modification of target data is evidenced in the supplied material. The Nmap output shows host discovery only; it does not show a port scan or service/version enumeration.

The WHOIS and DNS queries are information lookups; the HTTP and WAF tools generated network requests. The supplied material does not establish permission from the target owner.

3. Tools Used

Tool	Purpose
WHOIS	Retrieve publicly available domain-registration and registrar information.
WhatWeb	Identify observable web technologies and response characteristics from a website.
nslookup	Query DNS to resolve a domain name to IPv4 and IPv6 addresses.
cURL (-i)	Request a URL and display the HTTP response headers together with the response body.
Wafw00f	Use generic fingerprinting requests to check whether a web application firewall can be identified.
DNSRecon	Enumerate publicly published DNS records, including SOA, NS, MX, A, AAAA, TXT, DMARC, and SRV records.
Nmap (Zenmap interface)	Perform basic host discovery and report whether the selected target host responds; this run did not enumerate ports.

4. Activities Performed

4.1 WHOIS – Domain Registration Lookup

Command: whois google.com

The WHOIS output identified GOOGLE.COM as the queried domain and MarkMonitor Inc. as the registrar. The record showed a creation date of 15 September 1997, registry expiry date of 14 September 2028, and four Google name servers (NS1.GOOGLE.COM through NS4.GOOGLE.COM). The displayed registrant contact details were redacted for privacy. The WHOIS response also listed several client/server status restrictions and reported DNSSEC as unsigned.

Security relevance: The result provides basic registration and DNS delegation context. Registration data is informational and does not, by itself, indicate a security weakness. The DNSSEC field should be interpreted as the status reported by this WHOIS response at the time of collection.

4.2 WhatWeb – Web Technology Fingerprinting

Command: whatweb google.com

WhatWeb reported HTTP 301 redirects for the bare domain and successful HTTP 200 responses for the www.google.com destinations. The output identified the web server as gws, displayed Google page titles, reported HTML5 and cookie indicators, and listed response IP addresses including 142.250.140.101, 142.250.140.113, 142.251.154.119, and 142.251.151.119.

Security relevance: The observed redirects indicate that the requested hostnames direct browsers to the www host. Technology fingerprinting can help defenders understand the externally visible application surface. The detected indicators are observations, not proof of a vulnerability.

4.3 nslookup – DNS Resolution

Command: nslookup google.com

The query used DNS server 8.8.8.8. The non-authoritative answer returned multiple IPv4 addresses (142.250.129.100, 142.250.129.139, 142.250.129.101, 142.250.129.113, 142.250.129.102, and 142.250.129.138) and IPv6 addresses (2a00:1450:4009:c15::8a, ::65, ::64, and ::8b).

Security relevance: Multiple DNS answers can be used for traffic distribution, resilience, and regional routing. DNS results can vary over time and by resolver location; they should be recorded with the query date and resolver.

4.4 cURL – HTTP Response Header Inspection

Command: curl -i google.com

The response shown was HTTP/1.1 301 Moved Permanently with a Location header pointing to http://www.google.com/. The output also displayed Content-Type, Content-Security-Policy-Report-Only, Date, Expires, Cache-Control, Server: gws, Content-Length, X-XSS-Protection, and X-Frame-Options headers, followed by a short HTML redirect body.

Security relevance: The 301 status indicates a permanent redirect from the requested URL to the Location destination. The Content-Security-Policy-Report-Only header reports policy violations rather than enforcing the policy. Header presence or absence alone is not sufficient to conclude that a site is vulnerable.

4.5 Wafw00f – Generic WAF Fingerprinting

Command: wafw00f google.com

Wafw00f v2.4.2 checked https://google.com and returned: “No WAF detected by the generic detection.” The tool reported that it made seven requests.

Security relevance: This result means that Wafw00f did not identify a WAF using its generic detection method during this run. It does not prove that no WAF, filtering layer, or other protective service exists. Detection can be limited by the tool, its signatures, and the target's configuration.

4.6 DNSRecon – DNS Record Enumeration

Command: dnsrecon -d google.com

DNSRecon completed its general enumeration and reported SOA information for ns1.google.com; four authoritative name servers (ns1 through ns4.google.com); MX records pointing to smtp.google.com; multiple IPv4 (A) and IPv6 (AAAA) addresses; TXT records including domain/service verification entries and an SPF policy; a DMARC record at

_dmarc.google.com with policy p=reject; and SRV records for LDAP, CalDAVS, and CardDAVS services. The output also displayed an error stating that there was no answer for its DNSSEC query. The tool ended with “12 Records Found” and “Completed enumeration.” The full verification-token values are intentionally not reproduced in this report.

Security relevance: DNSRecon provides a broader view of publicly published DNS configuration than a basic address lookup. The discovered service records and mail-authentication policies are useful inventory information for an authorised domain owner. The DNSSEC query error is a query outcome, not proof of a DNSSEC vulnerability; it should be checked independently with the domain owner’s approved DNS tooling. Public TXT verification strings should not be treated as passwords or secrets, but need not be copied into a general report.

4.7 Nmap / Zenmap – Basic Host Discovery

Command / scan: Nmap 7.991, launched through the Zenmap graphical interface; target entered: google.com.

Observed output: Nmap reported that google.com resolved to two IP addresses and selected 142.251.208.142 for this scan. The selected host was reported as up. The output also listed the IPv6 address 2a00:1450:4006:821::200e as another address that was not scanned, and gave the reverse-DNS name ncmrsb-ao-in-f14.1e100.net. The scan completed in 0.34 seconds.

Interpretation and limitation: “Host is up” indicates that the selected address responded to Nmap's host-discovery method at the time of the test. This output does not contain an open-port list, service fingerprints, operating-system identification, or vulnerability results. The IPv6 address was not tested. Therefore, this result must not be presented as evidence that the host has no open ports or vulnerabilities.

Security relevance: Basic host discovery is useful for confirming reachability within an approved assessment scope. Any follow-on port or service testing must be explicitly covered by written permission, target-owner rules, and agreed rate limits.

5. Risk Analysis / Impact

The table below records observations from the supplied outputs. These are not confirmed vulnerabilities.

Observation	Evidence	Assessment	Interpretation
WHOIS registration and name-server data visible	Public registration metadata and DNS delegation information were returned.	Low	Expected public information; not a confirmed vulnerability.
Multiple DNS addresses returned	The resolver returned several IPv4 and IPv6 addresses.	Informational	Normal DNS behaviour; no weakness established.
HTTP redirect observed	The bare domain returned HTTP 301 and a Location header.	Informational	Redirect behaviour is not itself a vulnerability.
Security-related headers observed	The response included CSP-Report-Only, X-Frame-Options and X-XSS-Protection fields.	Informational	Header values require context and policy review; no exploitability was tested.
WAF not identified by generic detection	Wafw00f did not detect a WAF in this run.	Unconfirmed	A negative fingerprinting result is not proof that protective controls are absent.
DNS records enumerated	DNSRecon returned SOA, NS, MX, A, AAAA, TXT, DMARC, and SRV information; DNSSEC query returned no answer.	Informational	Useful DNS inventory. The DNSSEC query error alone does not establish a security weakness.
Target host responded to basic discovery	Nmap 7.991 reported host 142.251.208.142 as up; one additional IPv6 address was listed but not scanned.	Informational	Confirms reachability of the selected address at scan time only. No ports, services, or vulnerabilities were assessed.

6. Recommendations

Keep all further testing within a documented, written authorisation scope that names the exact domains, IP ranges, methods, dates, and rate limits permitted.

Treat WHOIS, DNS, technology-fingerprinting, and HTTP-header outputs as reconnaissance observations rather than confirmed vulnerabilities.

For systems owned by the organisation being assessed, review DNS records and registrar security settings periodically.

Review HTTP response headers against the organisation's intended security policy; validate configuration in an authorised test environment before making changes.

Do not infer the absence of a WAF from a single Wafw00f result. Confirm protective controls through approved documentation and owner-authorised testing.

Preserve timestamps, exact commands, tool versions, and unedited evidence screenshots so findings can be reproduced.

7. Conclusion

The exercise introduced core reconnaissance concepts using seven common tools. WHOIS provided registration details; WhatWeb returned visible technology and response indicators; nslookup resolved the domain to multiple IPv4 and IPv6 addresses; cURL displayed an HTTP 301 redirect and response headers; Wafw00f did not identify a WAF through its generic detection method; DNSRecon enumerated SOA, NS, MX, A, AAAA, TXT, DMARC, and SRV records; and Nmap reported that the selected IPv4 host was up. These outputs describe publicly observable behaviour and do not confirm a vulnerability. The work should be treated as an educational reconnaissance record. Any additional testing of google.com requires explicit authorisation from the system owner and compliance with the approved scope.

8. Evidence Collected

The following screenshots were supplied during the exercise. Captions identify the evidence type; the original terminal output remains the source for the exact values transcribed in this report.

WHOIS lookup – google.com

WhatWeb fingerprinting – google.com

DNS lookup – nslookup google.com

HTTP response inspection – curl -i google.com

Wafw00f evidence (terminal output transcribed): Wafw00f v2.4.2 checked https://google.com; generic detection reported “No WAF detected by the generic detection”; 7 requests were made. The text output was supplied in the conversation.

DNSRecon output – command and terminal text supplied in the conversation; no separate screenshot was provided.
Nmap / Zenmap evidence (terminal output supplied in the conversation): Nmap 7.991 reported google.com host 142.251.208.142 as up, listed one additional IPv6 address as not scanned, displayed reverse-DNS name ncmrsb-ao-in-f14.1e100.net, and completed in 0.34 seconds. The supplied output is a host-discovery result only.

9. Evidence and Submission Notes

Before submitting, compare this report with the instructor's official assignment brief. The assignment requirements or marking rubric were not included in the materials received. Also attach or reference written authorisation if the instructor expects a real client penetration test. Do not describe this reconnaissance exercise as an authorised penetration test unless that permission has actually been granted.

APPENDIX A — ADDITIONAL SCREENSHOT EVIDENCE

The following original screenshots were supplied on 13 November 2026 for inclusion in this report. The displayed terminal timestamps show that the underlying reconnaissance activities were conducted on 12 November 2026. Screenshots are retained as supporting evidence; interpretation remains limited to the outputs shown and does not establish authorisation or the presence/absence of vulnerabilities.

Figure 9. Wafw00f v2.4.2 generic fingerprinting output

Figure 10. DNSRecon enumeration output — SOA, NS, MX, A and AAAA records

Figure 11. DNSRecon enumeration output — TXT, DMARC and SRV records

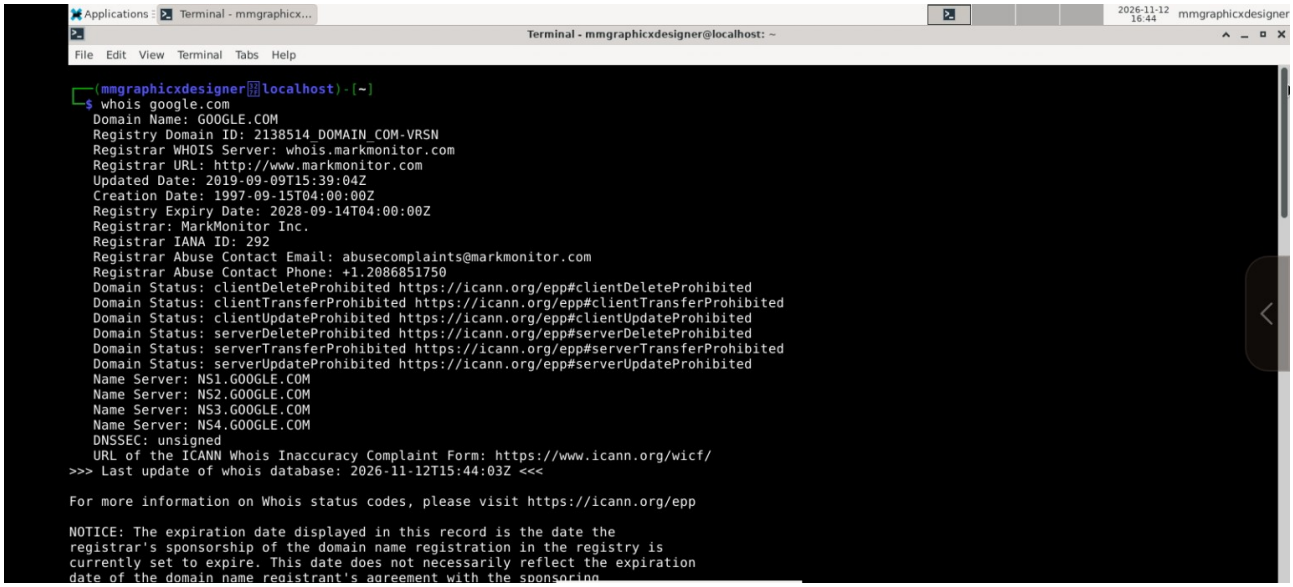
Figure 12. Zenmap topology view for the basic host-discovery scan

Figure 13. Zenmap/Nmap 7.991 host-discovery output

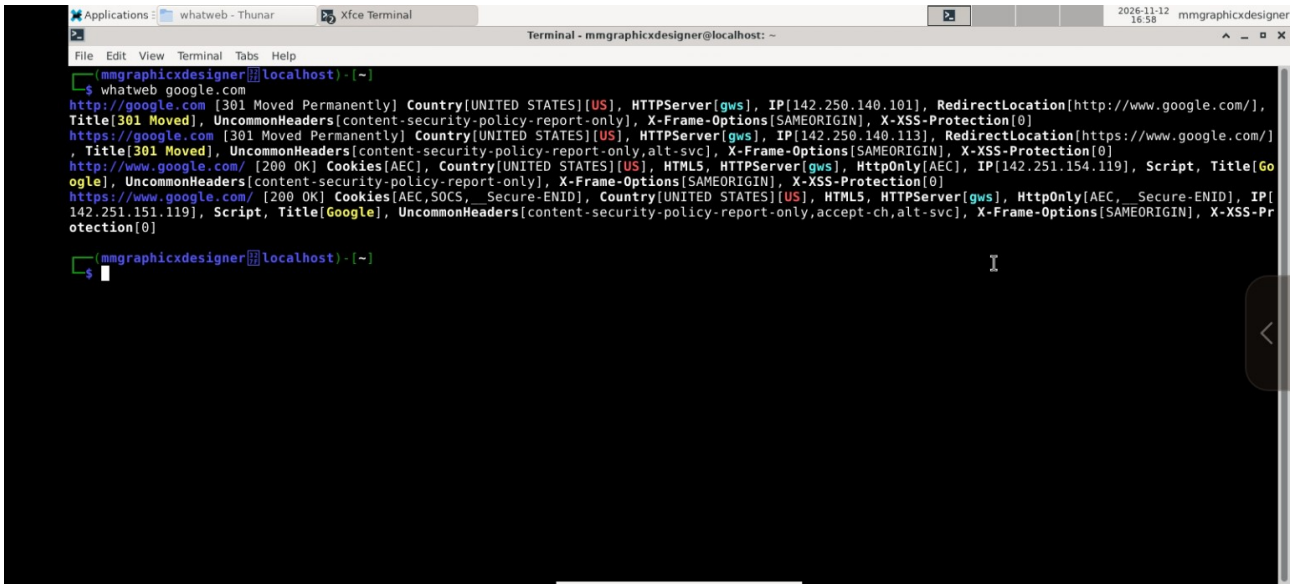
APPENDIX B — SCREENSHOT EVIDENCE

Previously embedded evidence images are retained first, followed by the additional screenshots supplied for this update. The terminal activity timestamps visible in the new screenshots correspond to 12 November 2026; this report issue date is 13 November 2026.

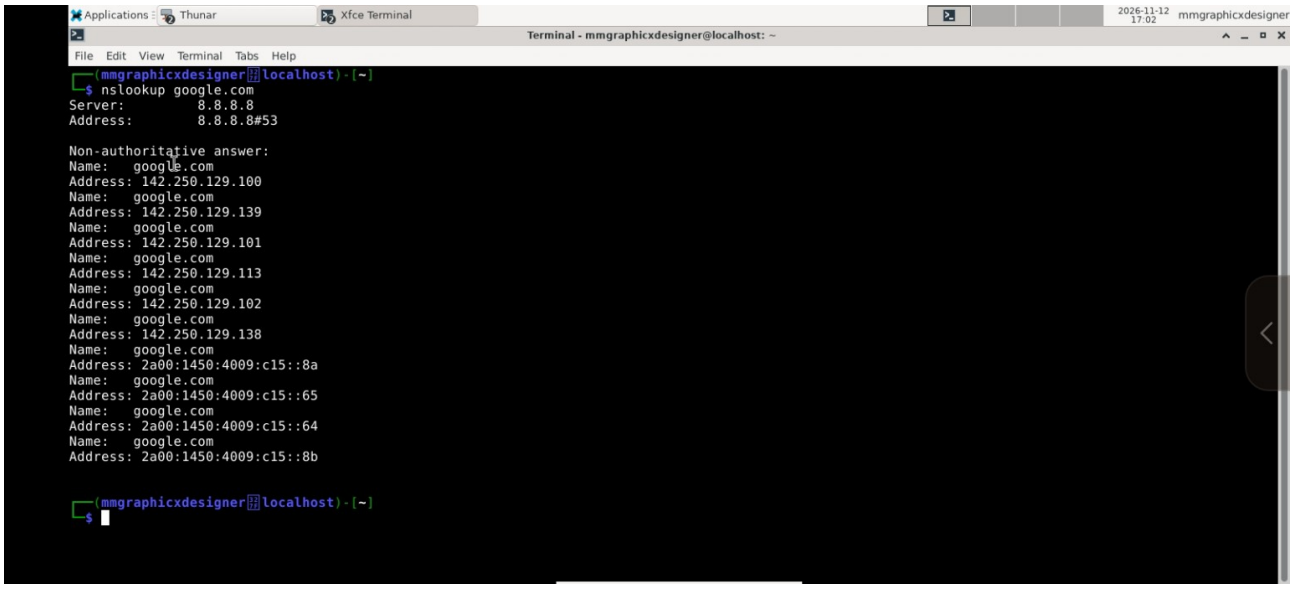
Previously included evidence image 1



Previously included evidence image 2



Previously included evidence image 3



Previously included evidence image 4

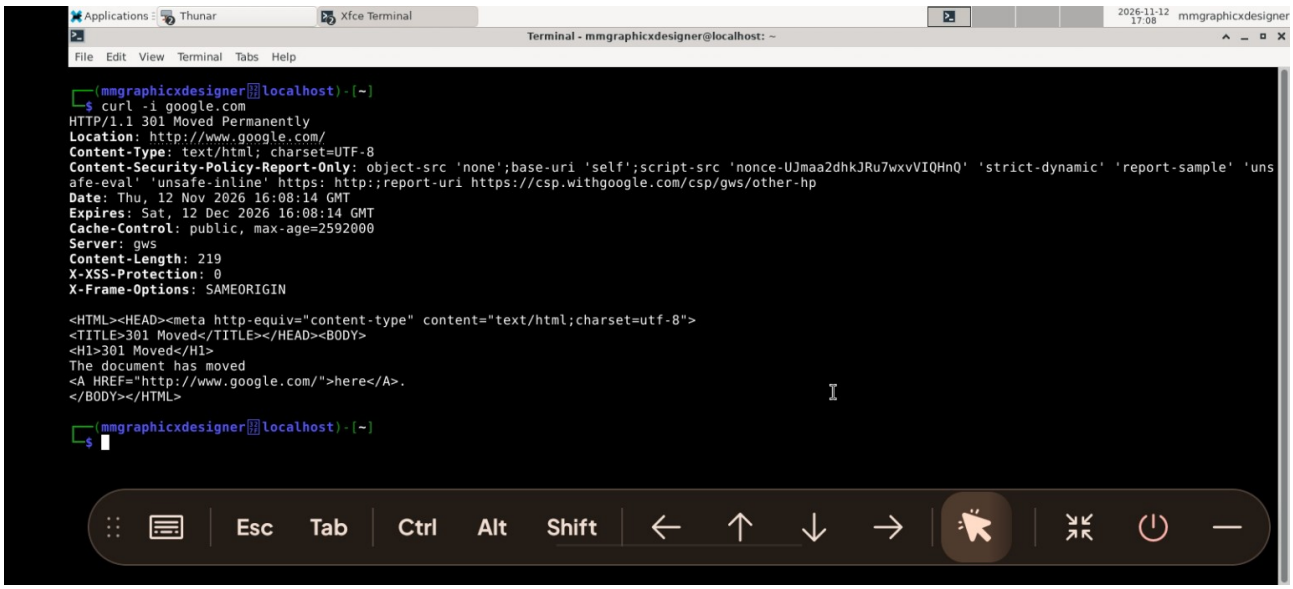


Figure 9. Wafw00f v2.4.2 generic fingerprinting output

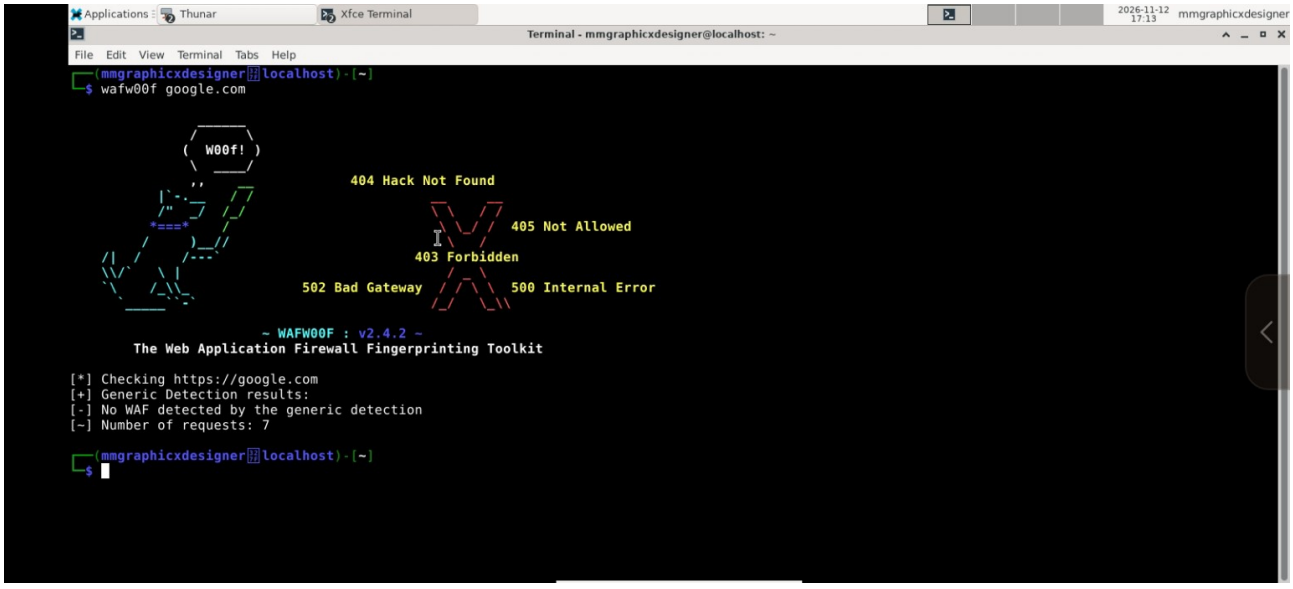


Figure 10. DNSRecon enumeration output — SOA, NS, MX, A and AAAA records

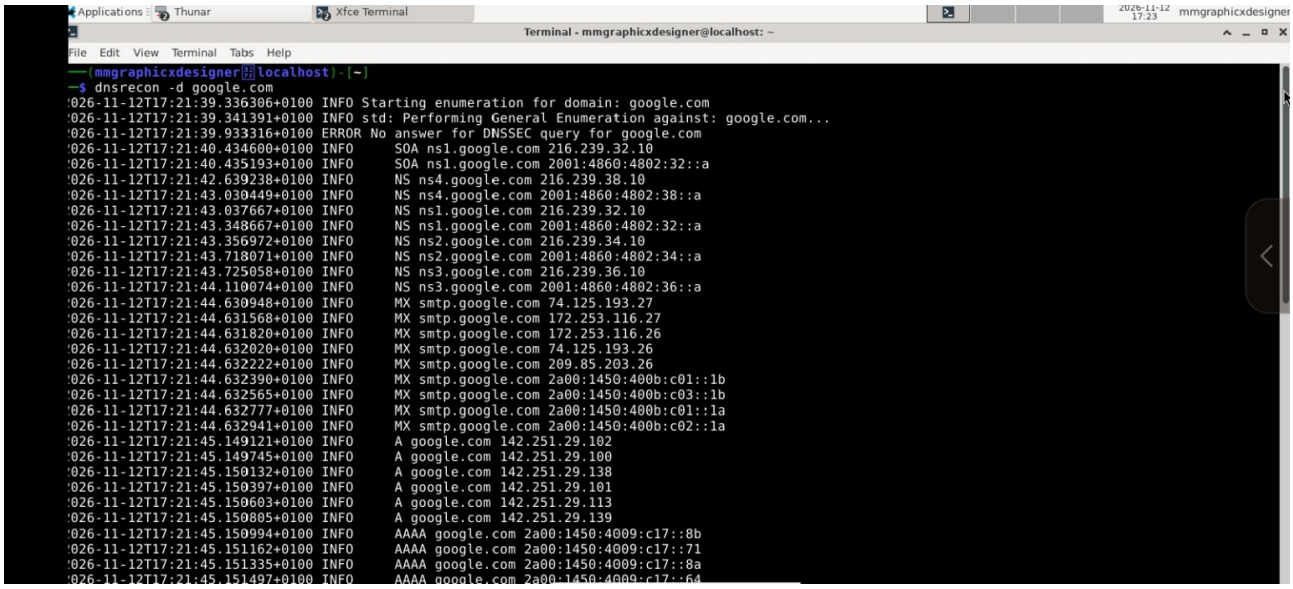


Figure 11. DNSRecon enumeration output — TXT, DMARC and SRV records

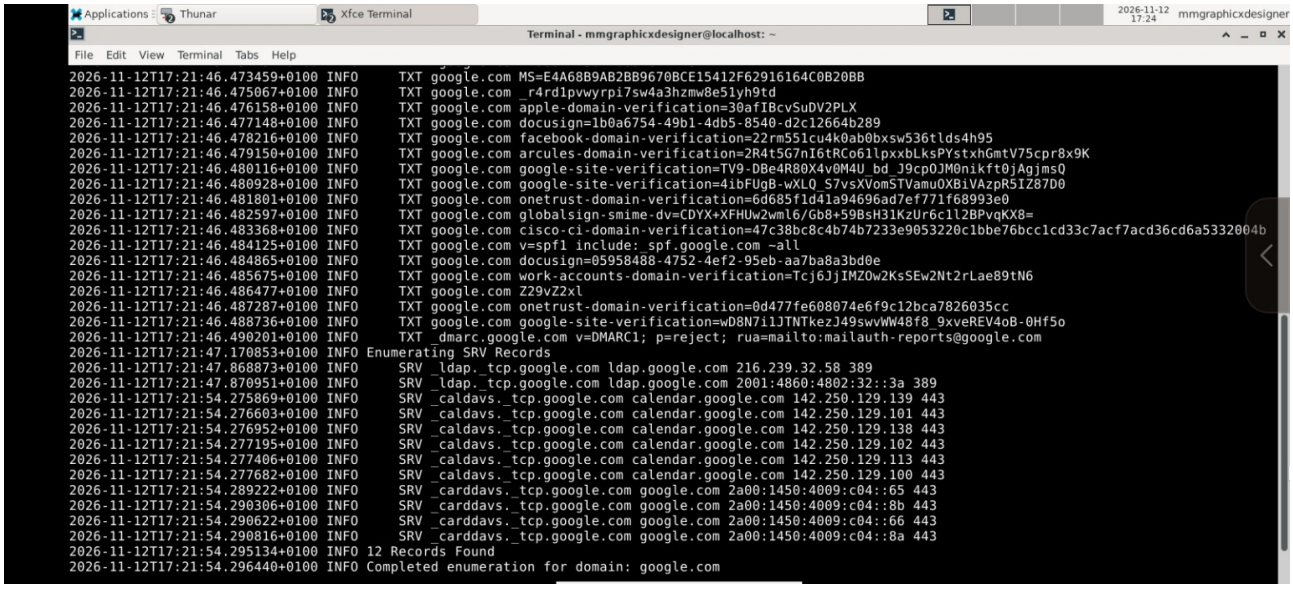


Figure 12. Zenmap topology view for the basic host-discovery scan

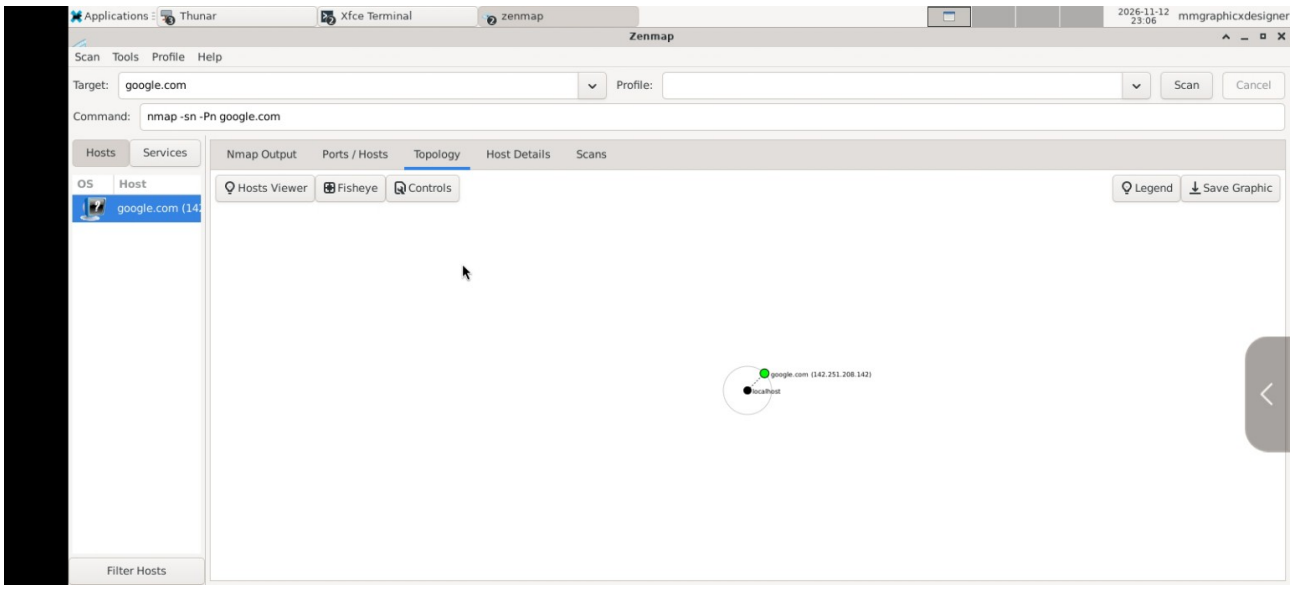


Figure 13. Zenmap/Nmap 7.991 host-discovery output

